

# Microsoft Access IV

## Switchboard

Similar to Java and C#, Microsoft Access is designed to allow you to create database-driven desktop applications.

Access applications are used extensively in large and medium-sized businesses for a variety of single user (and sometimes small workgroup) record-keeping tasks and are often used by small businesses for (among other things) accounting, invoicing, inventory control, pricing and general record-keeping. Many small computer consultancies have grown very successful developing Access applications and nothing else.

Most such applications consist of a collection of tables, a number of data-entry forms, and various reports. Many of these can be created with no programming at all, but when needed, programming can be done using a version of Visual Basic (called Visual Basic for Applications, or VBA) built into Access.

While most Windows applications – written in, for example, C# or Java – typically use menus and/or toolbars to access their functionality, Access encourages a slightly different approach: Rather than use menus or toolbars (though, if necessary, these can be created), Access is designed to use a “switchboard” to invoke the various forms and reports. A “switchboard” is simply a window that provides buttons to access the main forms and reports in an Access application.

This session will cover creating a “switchboard” form.

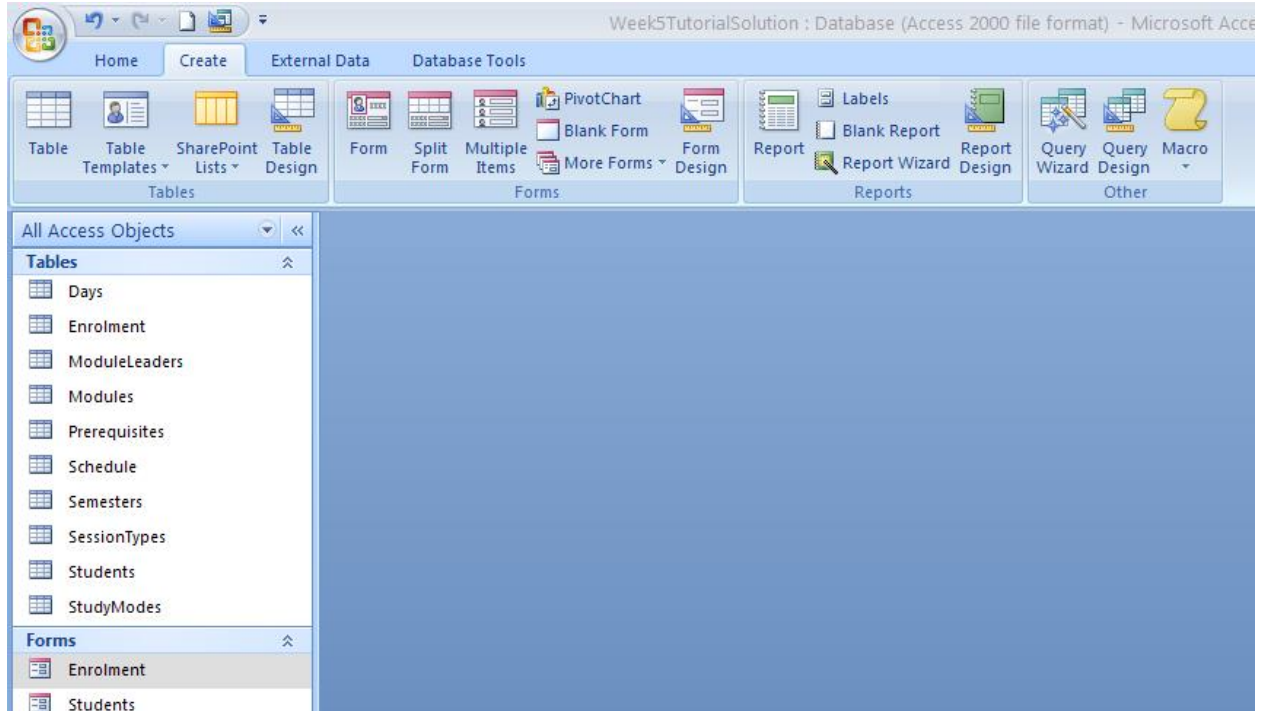
For this tutorial, you will use the same TutorialDatabase.mdb database that you used last week – which should be in the Tutorial02c folder of your portfolio – which by now should have forms to manipulate the data in all the tables.

We’ll start by creating a custom form that is not associated with any database table. Then we’ll add buttons to it that will allow your users to display all the appropriate forms.

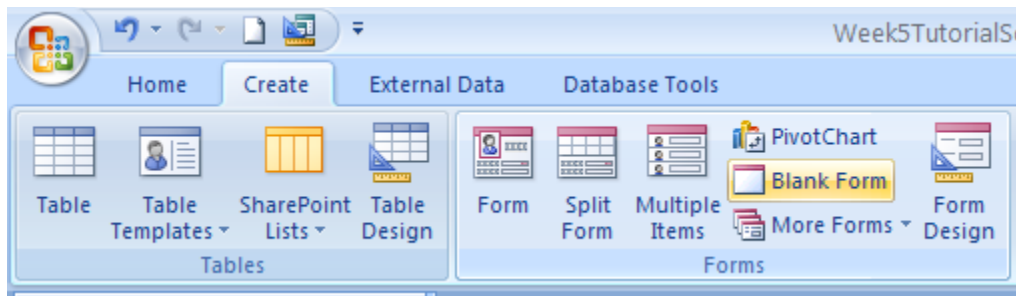
Load the .mdb (or .accdb) file into Access, and click on the Create tab.

(Note that the screens you see may be slightly different from those shown in this document, depending on the version of Access you use. However, the essential functionality will be the same.)

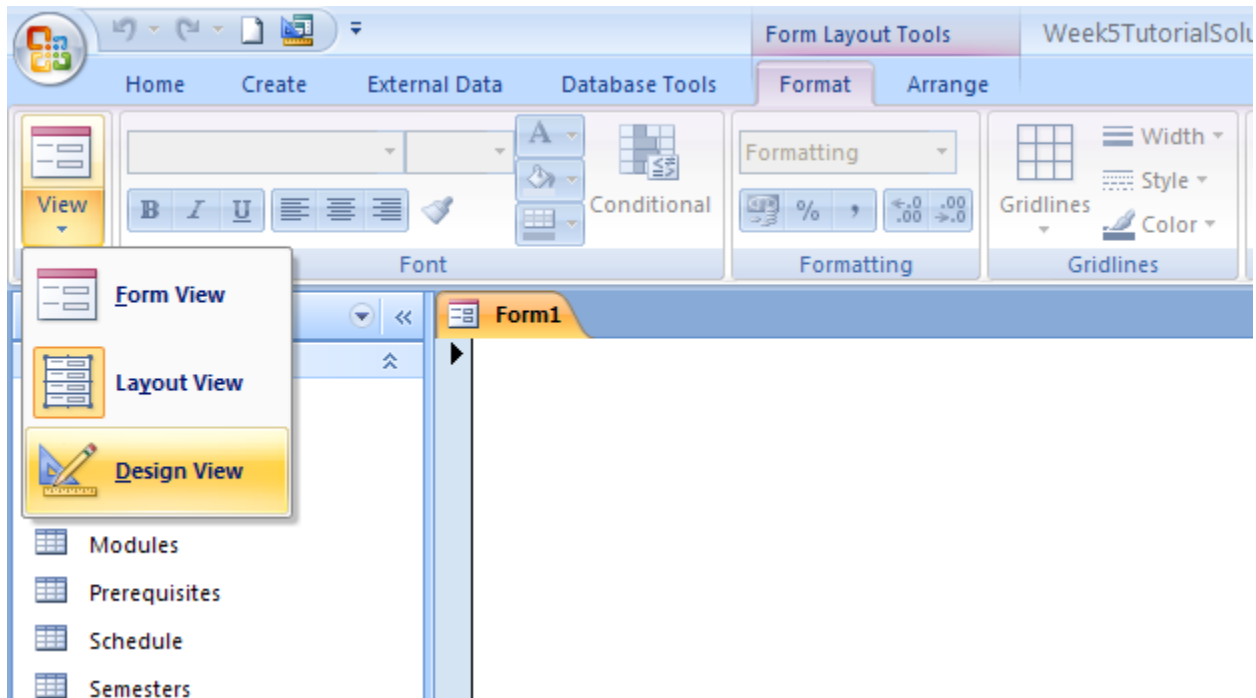
You should see something like the following:



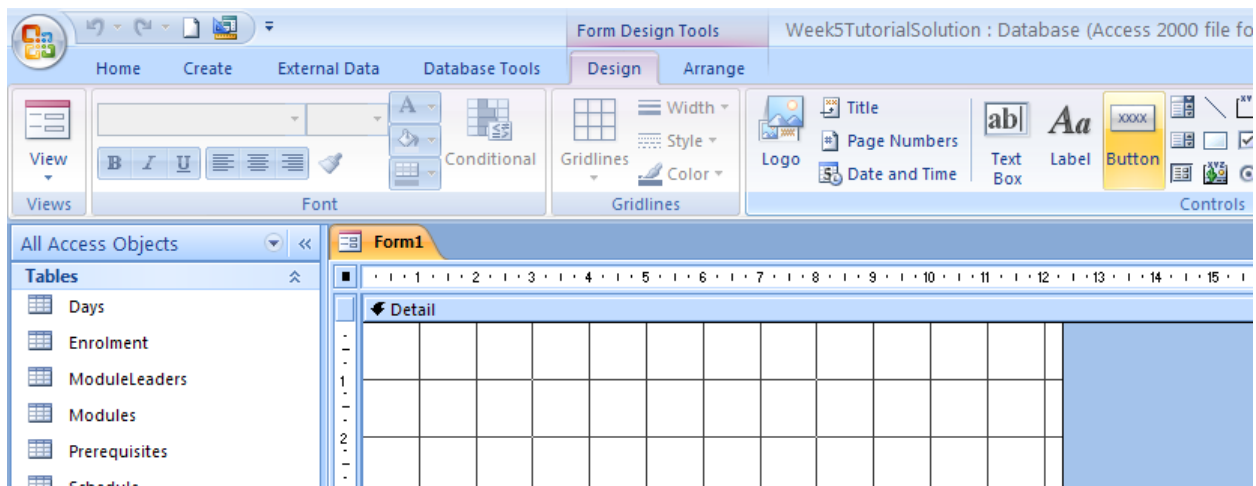
Now click on the “Blank Form” icon in the “Forms” panel:



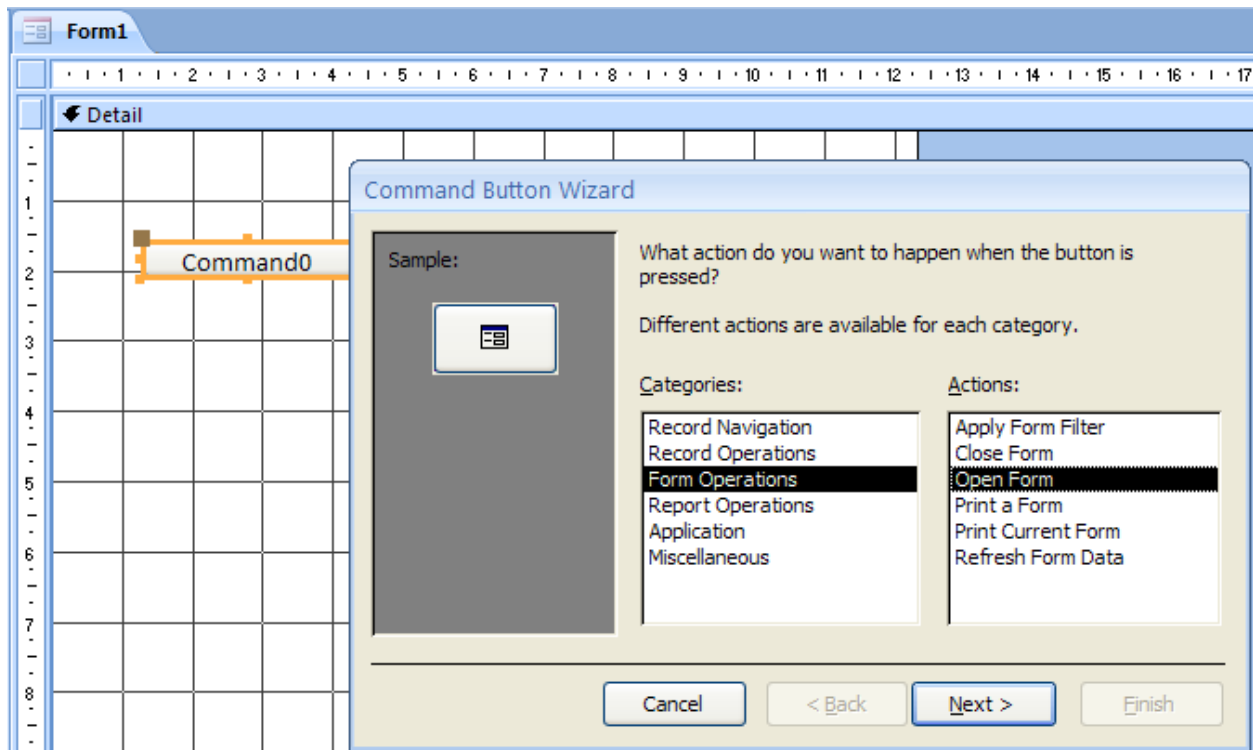
Switch the form into Design View:



Now we'll create a button that will be used to display one of the forms you created last week. Click on the "Button" icon in the "Controls" panel:



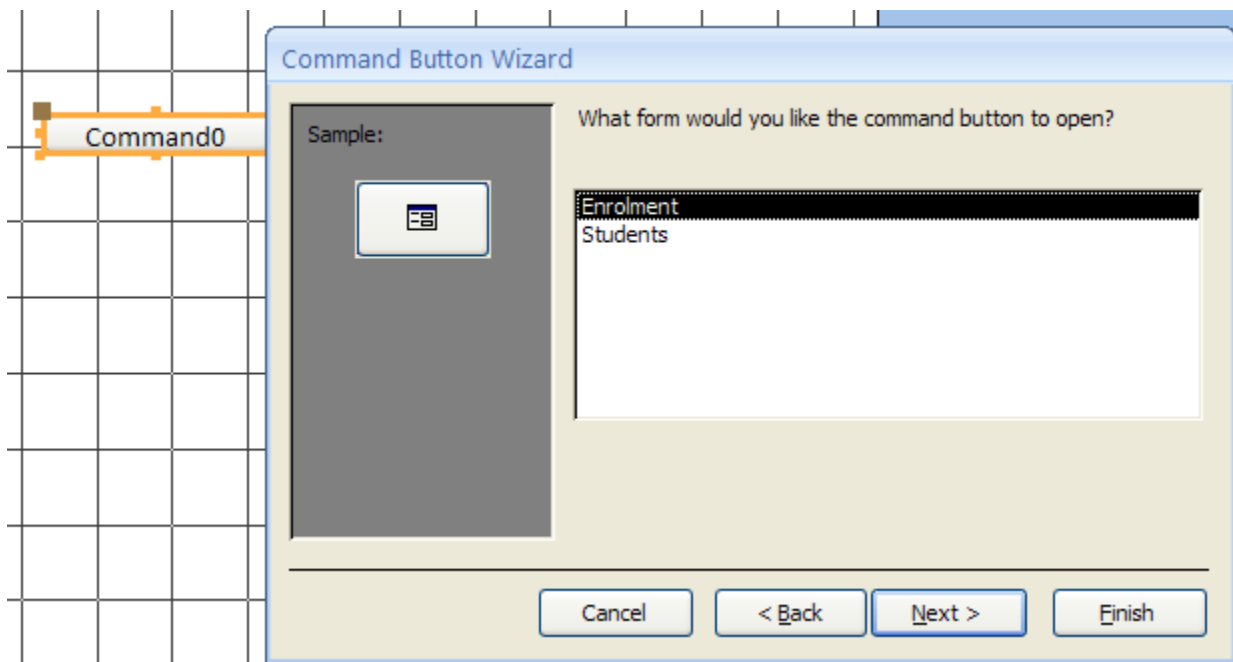
Click somewhere in the middle of the form canvas (the area that looks like graph paper) to start creating a button. The “Command Button Wizard” dialogue box should pop up:



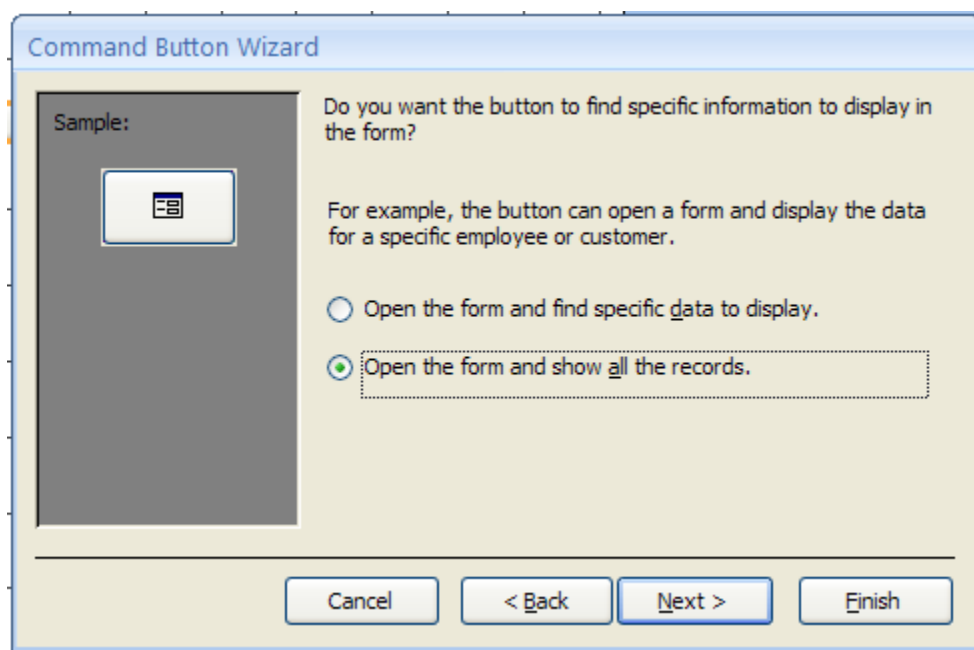
In the Categories list, select “Form Operations”, as shown above.

In the Actions list, select “Open Form”, as shown above.

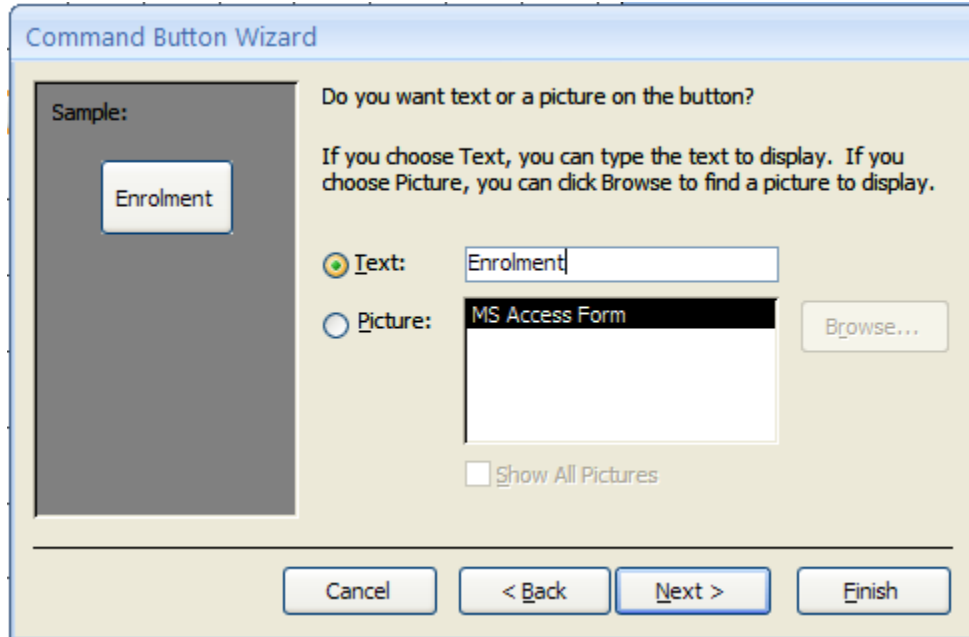
Then click the “Next” button. You should see something like the following:



Click on your Enrolment form, and press the “Next” button to display the following:



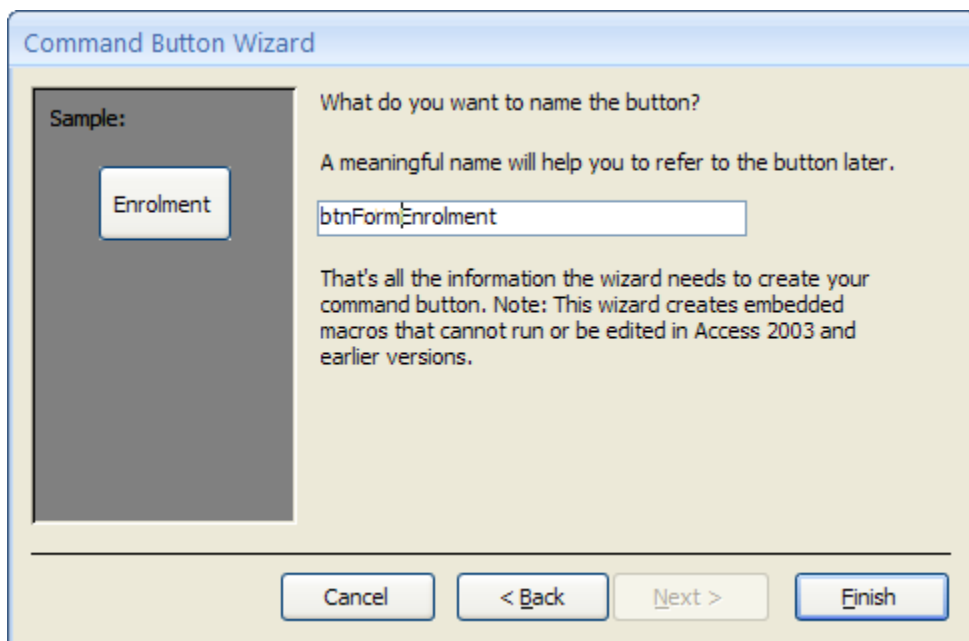
Select the option shown above. Then click “Next” to display the following:



The "Command Button Wizard" dialog box is shown at Step 1. On the left, a "Sample:" area displays a button labeled "Enrolment". The main text asks, "Do you want text or a picture on the button?" and provides instructions: "If you choose Text, you can type the text to display. If you choose Picture, you can click Browse to find a picture to display." The "Text:" radio button is selected, and the text "Enrolment" is entered in the adjacent text box. The "Picture:" radio button is unselected, with a list box showing "MS Access Form" and a "Browse..." button to its right. A "Show All Pictures" checkbox is also present. At the bottom are "Cancel", "< Back", "Next >", and "Finish" buttons.

Select the “Text” radiobutton, and type the name of the form into the associated text box. This will become the label that appears on the button.

Then click “Next”:



The "Command Button Wizard" dialog box is shown at Step 2. The "Sample:" area still shows the "Enrolment" button. The main text asks, "What do you want to name the button?" and explains, "A meaningful name will help you to refer to the button later." A text box contains the name "btnFormEnrolment". Below this, a note states: "That's all the information the wizard needs to create your command button. Note: This wizard creates embedded macros that cannot run or be edited in Access 2003 and earlier versions." The bottom buttons are "Cancel", "< Back", "Next >", and "Finish".

Give the button a name. This name won't be used here, but is useful when you create more complex applications with code written in VBA. It's good to get in the habit of appropriately naming buttons and other controls, however.

Click “Finish”. You should see a form similar to the following:

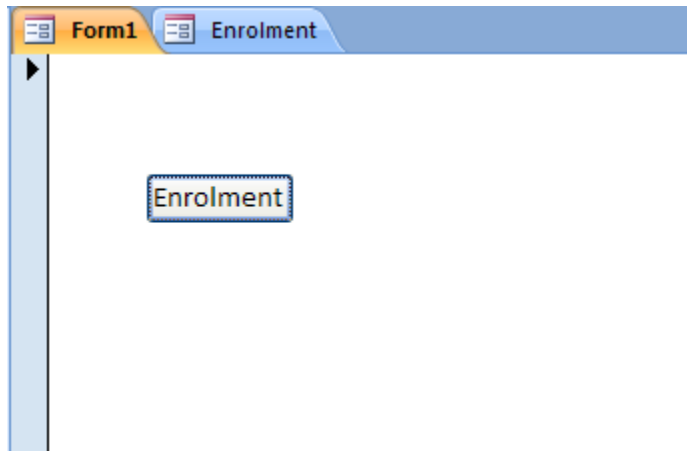
The screenshot shows a form design interface. At the top, there are tabs for 'Font' and 'Gridlines'. Below these is a tab labeled 'Form1'. A horizontal ruler at the top indicates columns 1 through 13. A vertical ruler on the left indicates rows 1 through 5. The main area is a grid. In the second row and second column, there is a button labeled 'Enrolment' with a small square icon to its left. The button has a blue border and a light blue fill.

Now switch back to Form View to try your form:

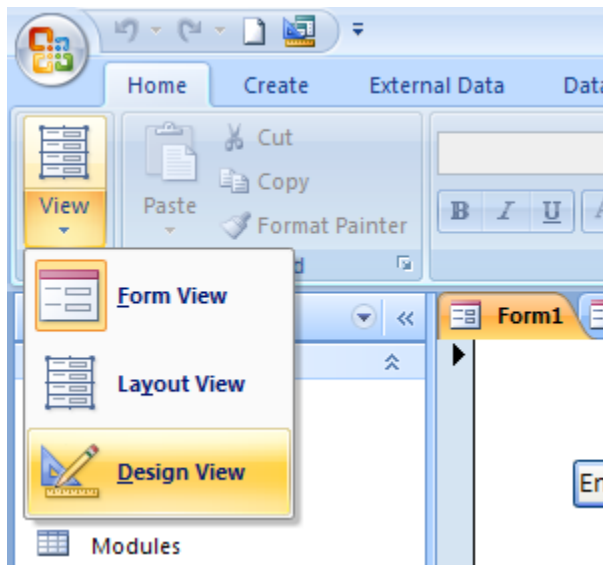
The screenshot shows the same form design interface as before, but with the 'View' menu open. The menu has three options: 'Form View' (selected), 'Layout View', and 'Design View'. Below the menu, there are tabs for 'Modules', 'Prerequisites', and 'Schedule'. The 'Form1' tab is still selected, and the 'Enrolment' button is visible in the grid.

Click on your new “Enrolment” button to see if it works. It should bring up your Enrolment form in a separate tab.

Now click back on the Form1 tab:

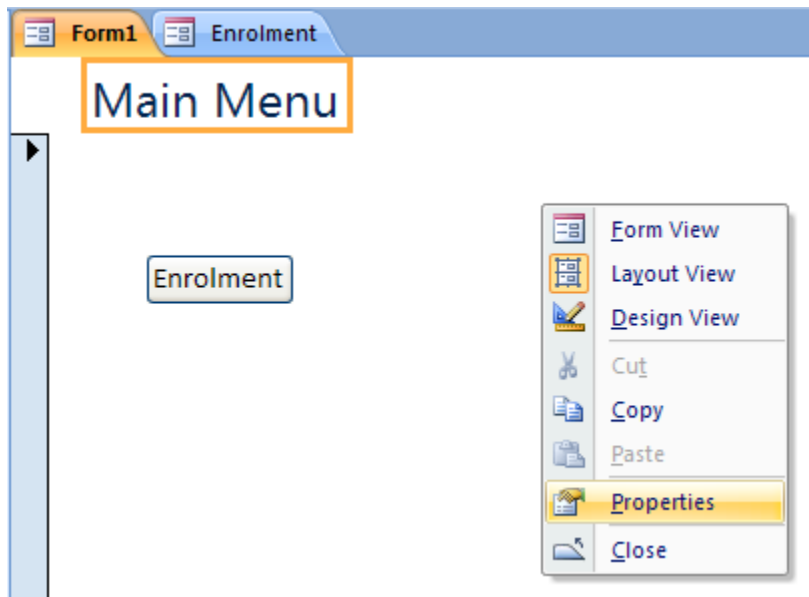


Go into Layout View:

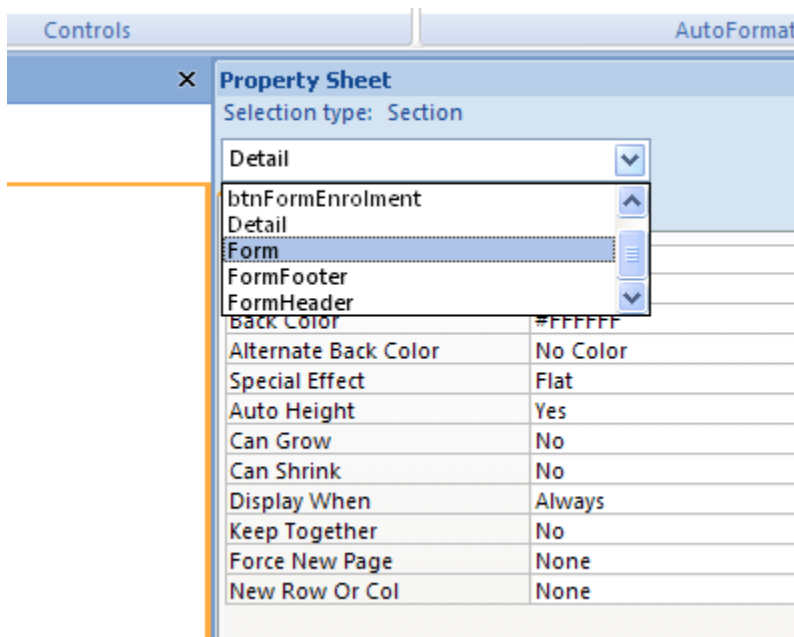




Now you can move the button to its final position, set the title on the form, and change the form's properties by right-clicking:



Most of the properties you'll want to set are in the "Form" section, as shown below:



For example, turn off (set to “No”) the record selectors and navigation bar. These are used when editing table contents, but aren’t required (and may be confusing to your users) on a Switchboard form:

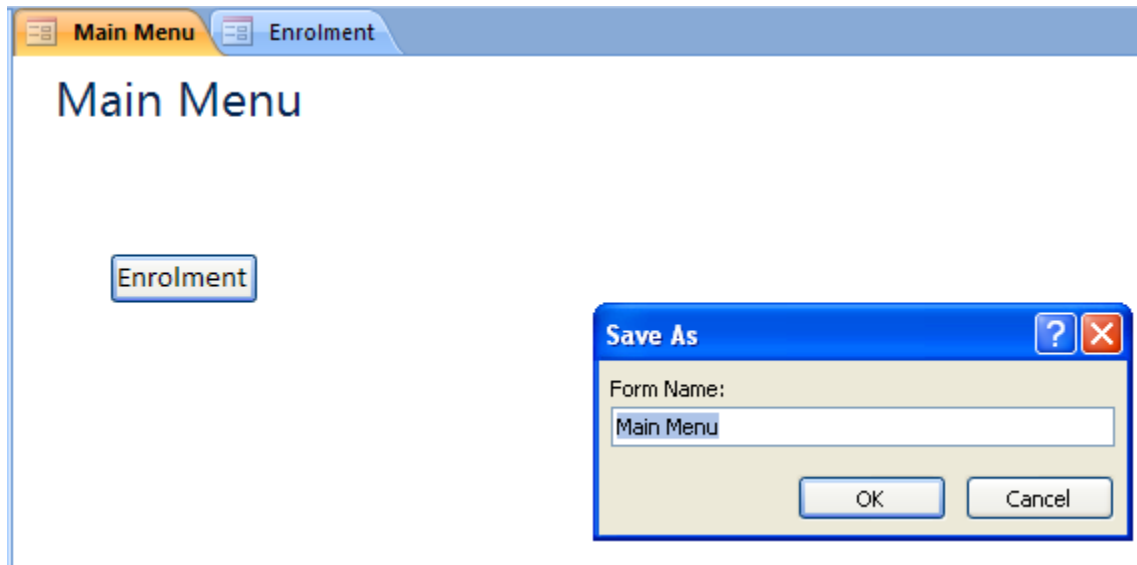
**Property Sheet** Selection type: Form

Form

Format Data Event Other All

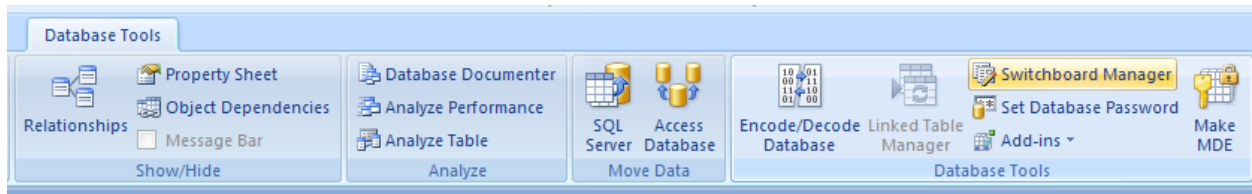
|                       |             |
|-----------------------|-------------|
| Caption               | Main Menu   |
| Default View          | Single Form |
| Allow Form View       | Yes         |
| Allow Datasheet View  | No          |
| Allow PivotTable View | No          |
| Allow PivotChart View | No          |
| Allow Layout View     | Yes         |
| Picture               | (none)      |
| Picture Tiling        | No          |
| Picture Alignment     | Center      |
| Picture Type          | Embedded    |
| Picture Size Mode     | Clip        |
| Width                 | 12.335cm    |
| Auto Center           | No          |
| Auto Resize           | Yes         |
| Fit to Screen         | Yes         |
| Border Style          | Sizable     |
| Record Selectors      | No          |
| Navigation Buttons    | No          |
| Navigation Caption    |             |
| Dividing Lines        | No          |
| Scroll Bars           | Both        |
| Control Box           | Yes         |
| Close Button          | Yes         |

When you're done, close your form and save it using a meaningful name, such as "Main Menu".

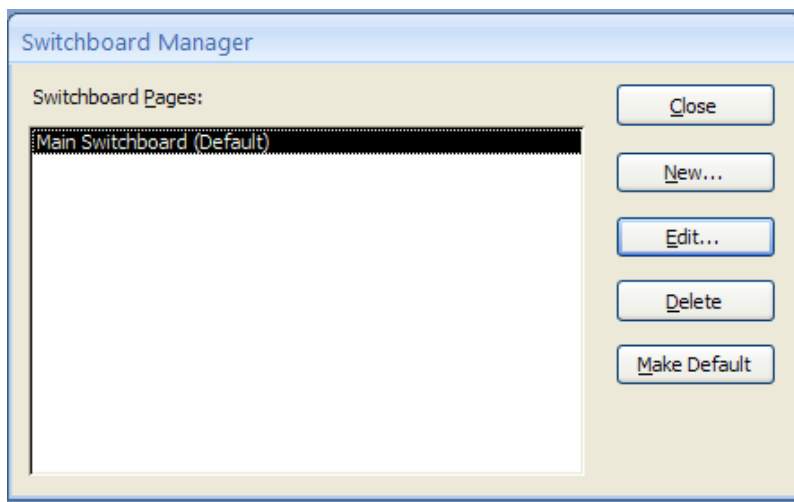


Once you've added buttons for all your other forms, you can use the "Main Menu" form (i.e., the switchboard) to conveniently load all your forms.

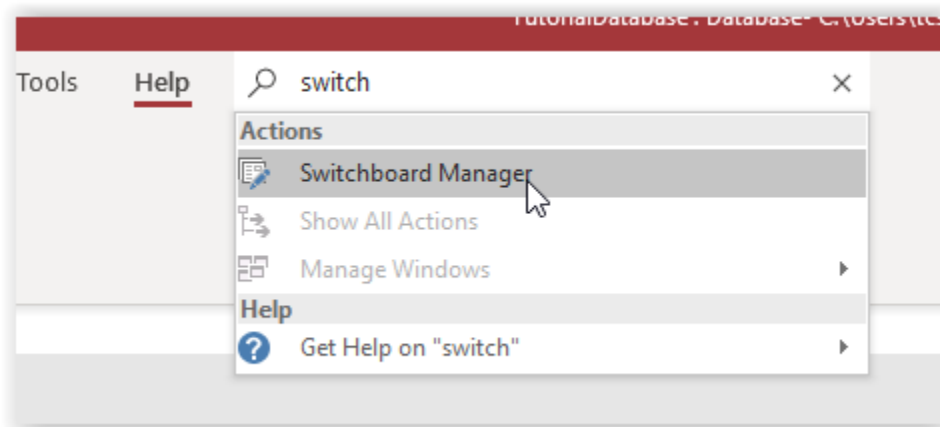
Because creating “Switchboard” forms is such a common activity, Access provides a mechanism to quickly create a Switchboard. You can access it by selecting the “Database Tools” tab and clicking on the “Switchboard Manager” icon in the “Database Tools” panel:



The “Switchboard Manager” dialogue box should appear.



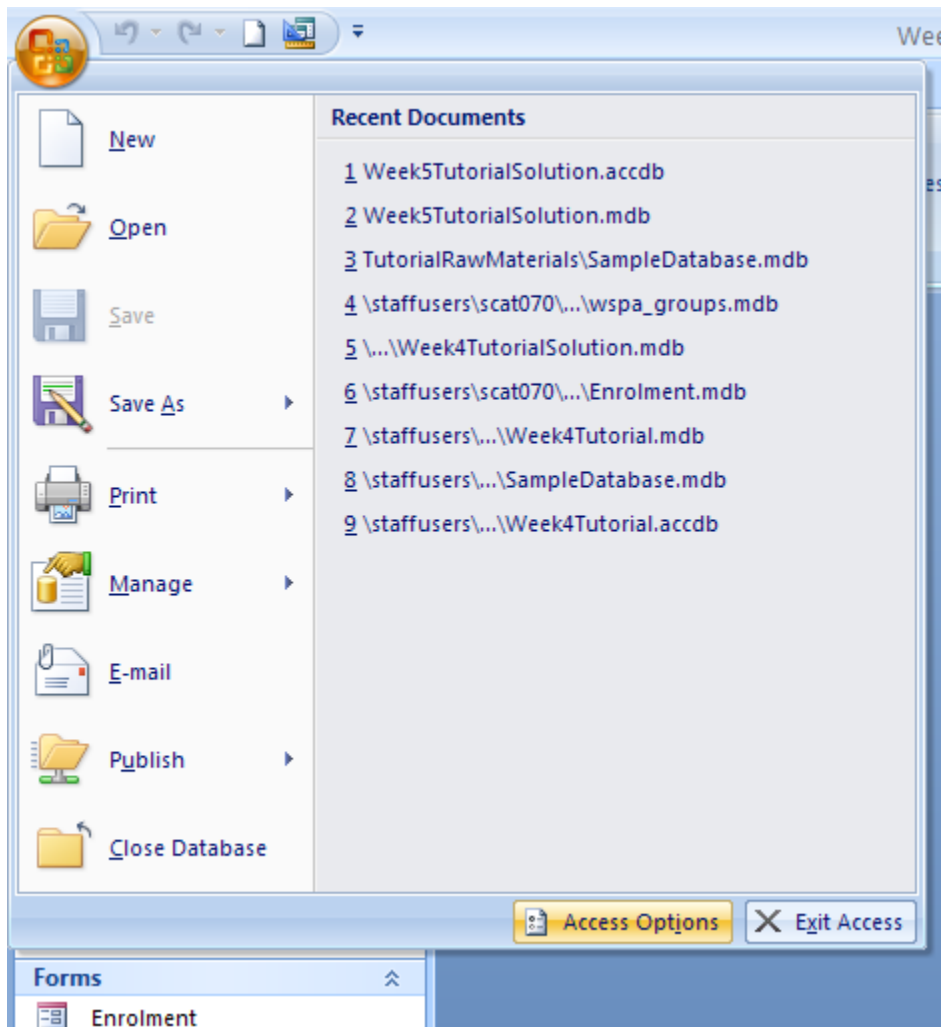
Or click on help, and search for Switchboard Manager:



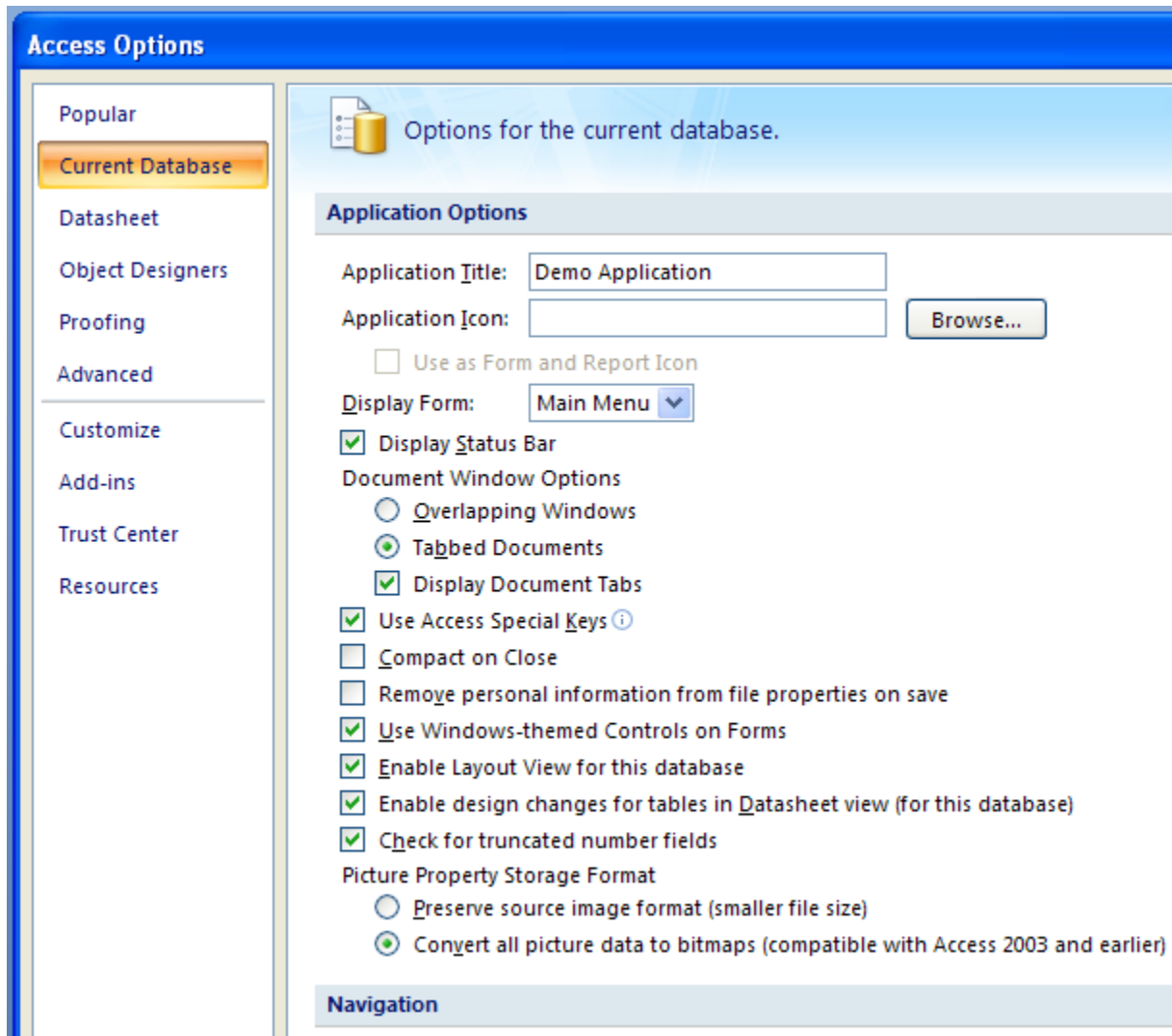
You may experiment with it if you like. Unfortunately, some Access developers have found it to be unreliable. If it works for you, great! If not, the steps described previously will always work, and require more effort than using the “Switchboard Manager”.

Now we'll configure Access to automatically display your Switchboard when your database is loaded.

Click on the main menu (the icon in the very upper left-hand corner of the Access window), and click the "Access Options" button, shown below:



You should now see the following:



You can fill in the name for your application in the “Application Title” blank.

You can select your Switchboard using the “Display form” combobox.

Other options may be selected as you see fit.

When you close “Access Options”, the following should appear:



Close and reopen your database. Your Switchboard should automatically appear.

**NOTE:**

**If you accidentally select a set of options that won't allow you to edit your database, exit from Access, then double-click on the database file whilst holding down the left-shift key. That will bypass the Switchboard.**

## **Task**

Create a switchboard for the database, with buttons to load your forms.